

EXPERIMENT – 4

Demonstration of Hash Cracking Using John the Ripper Tool in Kali Linux

Aim:

To demonstrate password hash cracking using the John the Ripper tool in Kali Linux for understanding password vulnerabilities and security analysis.

Software Requirements:

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Security Tool:** John the Ripper
- **System Requirements:**
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Introduction:

Password hashes are commonly used to store user credentials securely. However, weak passwords and outdated hashing techniques can be exploited using password-cracking tools.

John the Ripper is a popular open-source password auditing and cracking tool used to identify weak passwords by performing dictionary, brute-force, and hybrid attacks.

Using Kali Linux in a virtualized environment allows safe and ethical experimentation without affecting real systems.

Procedure:

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Open Terminal

1. Click on Terminal from the Kali Linux menu.
2. Ensure John the Ripper is installed:
`john --version`

Step 3: Prepare the Hash File

1. Create a text file containing password hashes (example: hash.txt).
2. Store the hash inside the file:
`nano hash.txt`
3. Save and exit the file.

Step 4: Run John the Ripper

1. Execute the John the Ripper command:
john hash.txt
2. The tool starts cracking the hash using default wordlists.

Step 5: View Cracked Passwords

1. To display cracked passwords:
john --show hash.txt

Output:

```
(root@kali)-[~]
# echo "482c811da5d5b4bc6d497ffa98491e38" > hash.txt

(root@kali)-[~]
# john --format=raw-md5 hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=6
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
Proceeding with incremental:ASCII
0g 0:00:00:49 3/3 0g/s 59739Kp/s 59739Kc/s 59739KC/s 8j0bf5..8j0hmo
0g 0:00:00:51 3/3 0g/s 59690Kp/s 59690Kc/s 59690KC/s eezhf-..eez62c
0g 0:00:00:56 3/3 0g/s 60172Kp/s 60172Kc/s 60172KC/s pbp5eb4..pbp7e8!
0g 0:00:00:58 3/3 0g/s 60414Kp/s 60414Kc/s 60414KC/s l0ser1171..l0sel2086
0g 0:00:00:59 3/3 0g/s 60494Kp/s 60494Kc/s 60494KC/s pipoksust..piphituft
0g 0:00:01:00 3/3 0g/s 60579Kp/s 60579Kc/s 60579KC/s eyokbes..eyokb58
0g 0:00:01:01 3/3 0g/s 60591Kp/s 60591Kc/s 60591KC/s d09jeaw..d09jebe
0g 0:00:01:02 3/3 0g/s 60586Kp/s 60586Kc/s 60586KC/s creakerment..saurtymn
0g 0:00:01:03 3/3 0g/s 60634Kp/s 60634Kc/s 60634KC/s krby2di1..krbyfacc
0g 0:00:01:04 3/3 0g/s 60653Kp/s 60653Kc/s 60653KC/s peatsecolf..peatstupe2
0g 0:00:01:19 3/3 0g/s 60913Kp/s 60913Kc/s 60913KC/s gum0tng..gum0rn3
0g 0:00:01:20 3/3 0g/s 60796Kp/s 60796Kc/s 60796KC/s clifulc2..clifjkir
0g 0:00:01:21 3/3 0g/s 60656Kp/s 60656Kc/s 60656KC/s 26s6tul7..26s6jd2s
0g 0:00:00:00 DONE (2026-04-22 12:28) 0g/s 50.00p/s 50.00c/s 50.00C/s hello12
3
Session completed.
```

Result:

Hash cracking was successfully demonstrated using the John the Ripper tool in Kali Linux.